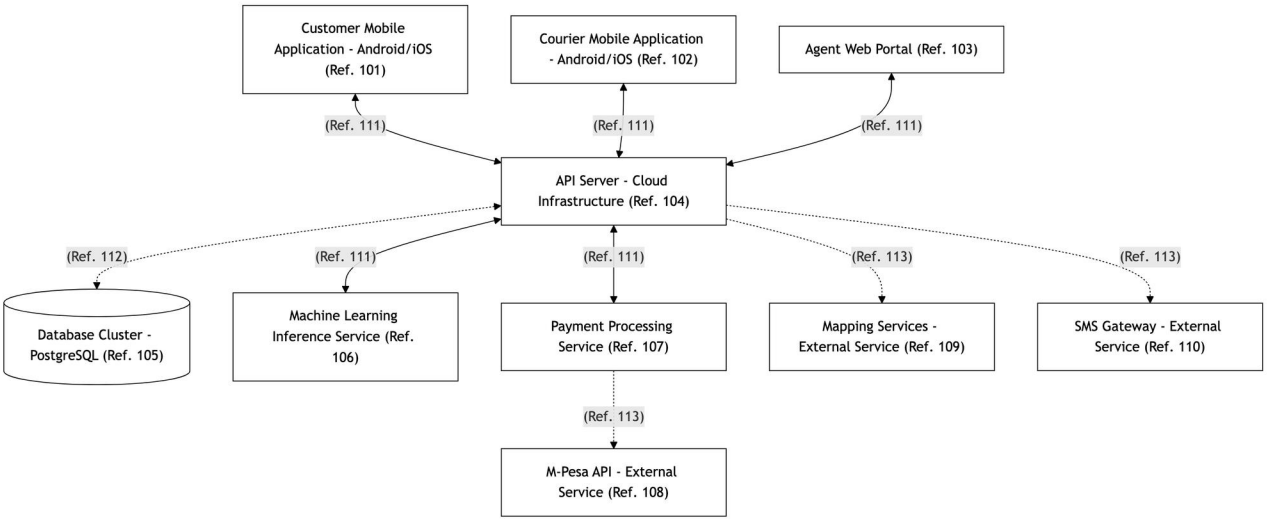
Customer Mobile Application - Android/iOS (Ref. 101)
Courier Mobile Application - Android/iOS (Ref. 102)
Agent Web Portal (Ref. 103)
(Ref. 111)
(Ref. 111)
(Ref. 111)
API Server - Cloud Infrastructure (Ref. 104)
(Ref. 112)
(Ref. 111)
(Ref. 111)
(Ref. 113)
(Ref. 113)
Database Cluster - PostgreSQL (Ref. 105)
Machine Learning Inference Service (Ref. 106)
Payment Processing Service (Ref. 107)
Mapping Services - External Service (Ref. 109)
SMS Gateway - External Service (Ref. 110)
(Ref. 113)
M-Pesa API - External Service (Ref. 108)

Fig. 1

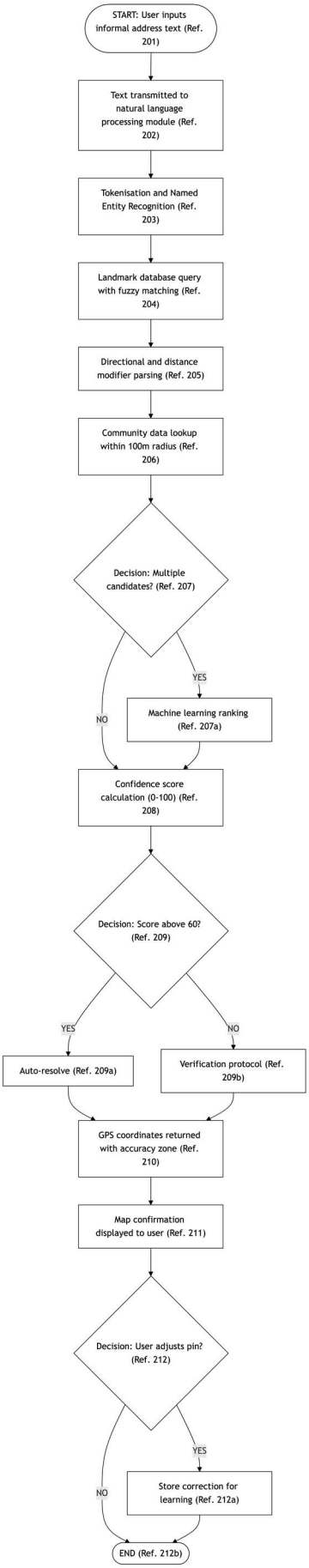
START: User inputs informal address text (Ref. 201)
Text transmitted to natural language processing module (Ref. 202)
Tokenisation and Named Entity Recognition (Ref. 203)
Landmark database query with fuzzy matching (Ref. 204)
Directional and distance modifier parsing (Ref. 205)
Community data lookup within 100m radius (Ref. 206)
Decision: Multiple candidates? (Ref. 207)
YES
NO
Machine learning ranking (Ref. 207a)
Confidence score calculation (0-100) (Ref. 208)
Decision: Score above 60? (Ref. 209)
YES
NO
Auto-resolve (Ref. 209a)
Verification protocol (Ref. 209b)
GPS coordinates returned with accuracy zone (Ref. 210)
Map confirmation displayed to user (Ref. 211)
Decision: User adjusts pin? (Ref. 212)
YES
NO
Store correction for learning (Ref. 212a)
END (Ref. 212b)

Fig. 2

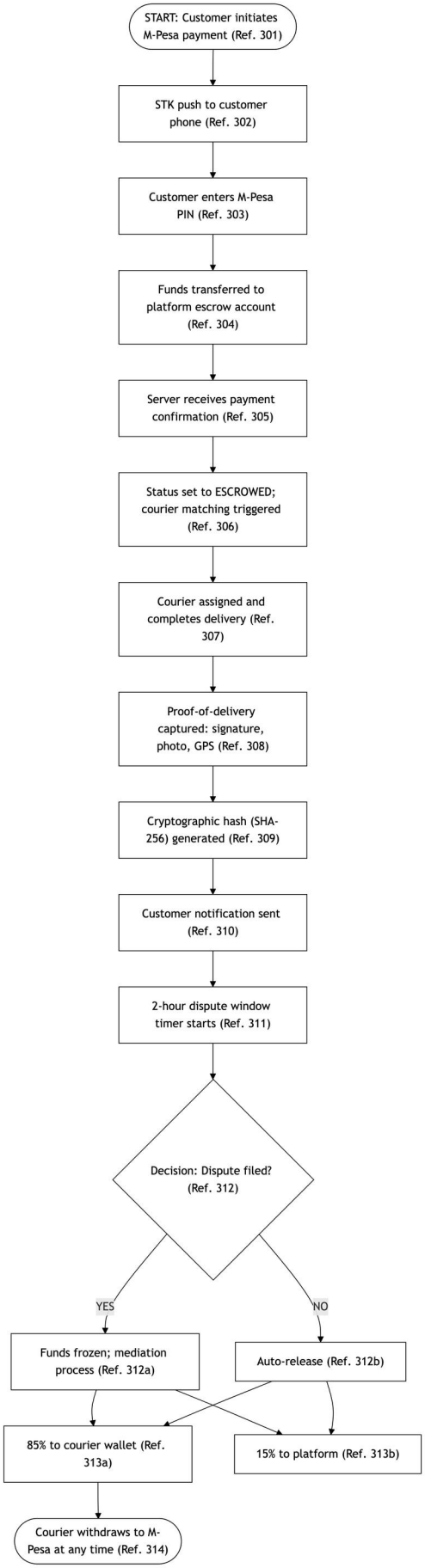
START: Customer initiates M-Pesa payment (Ref. 301)
STK push to customer phone (Ref. 302)
Customer enters M-Pesa PIN (Ref. 303)
Funds transferred to platform escrow account (Ref. 304)
Server receives payment confirmation (Ref. 305)
Status set to ESCROWED; courier matching triggered (Ref. 306)
Courier assigned and completes delivery (Ref. 307)
Proof-of-delivery captured: signature, photo, GPS (Ref. 308)
Cryptographic hash (SHA-256) generated (Ref. 309)
Customer notification sent (Ref. 310)
2-hour dispute window timer starts (Ref. 311)
Decision: Dispute filed? (Ref. 312)
YES
NO
Funds frozen; mediation process (Ref. 312a)
Auto-release (Ref. 312b)
85% to courier wallet (Ref. 313a)
15% to platform (Ref. 313b)
Courier withdraws to M-Pesa at any time (Ref. 314)

Fig. 3

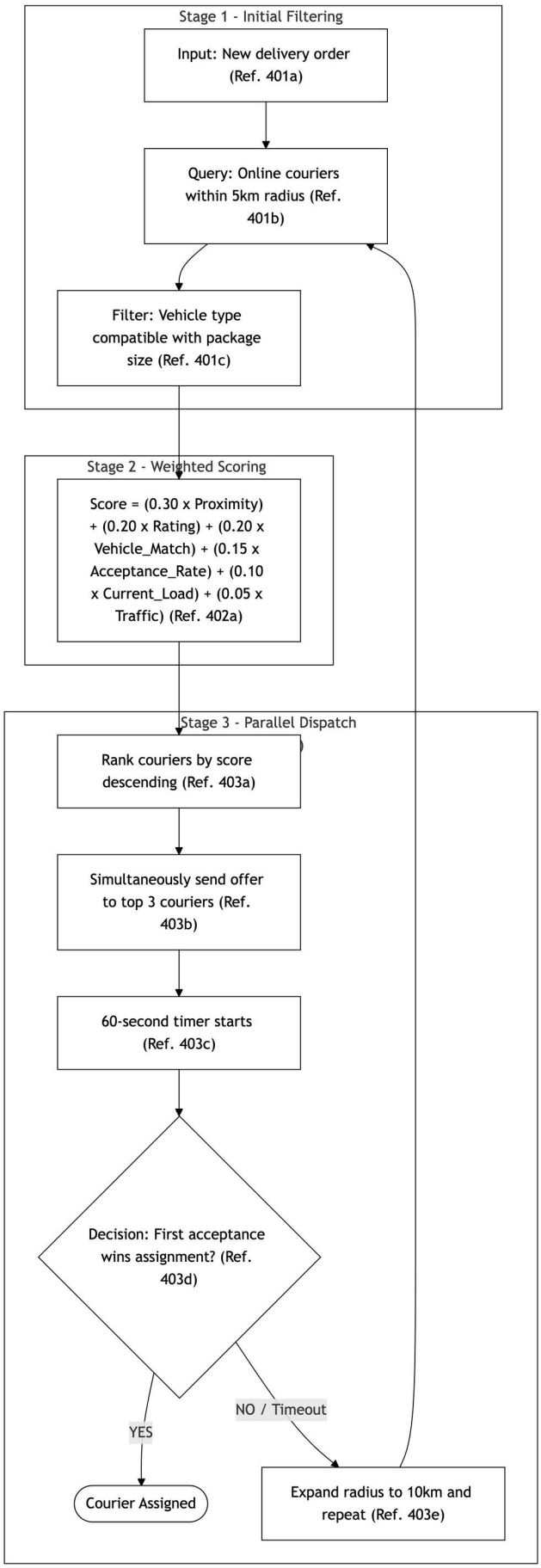
Stage 1 - Initial Filtering
Input: New delivery order (Ref. 401a)
Query: Online couriers within 5km radius (Ref. 401b)
Filter: Vehicle type compatible with package size (Ref. 401c)
Stage 2 - Weighted Scoring
Score = (0.30 x Proximity) + (0.20 x Rating) + (0.20 x Vehicle_Match) + (0.15 x Acceptance_Rate) + (0.10 x Current_Load) + (0.05 x Traffic) (Ref. 402a)
Stage 3 - Parallel Dispatch
Rank couriers by score descending (Ref. 403a)
Simultaneously send offer to top 3 couriers (Ref. 403b)
60-second timer starts (Ref. 403c)
Decision: First acceptance wins assignment? (Ref. 403d)
YES
NO / Timeout
Courier Assigned
Expand radius to 10km and repeat (Ref. 403e)

Fig. 4

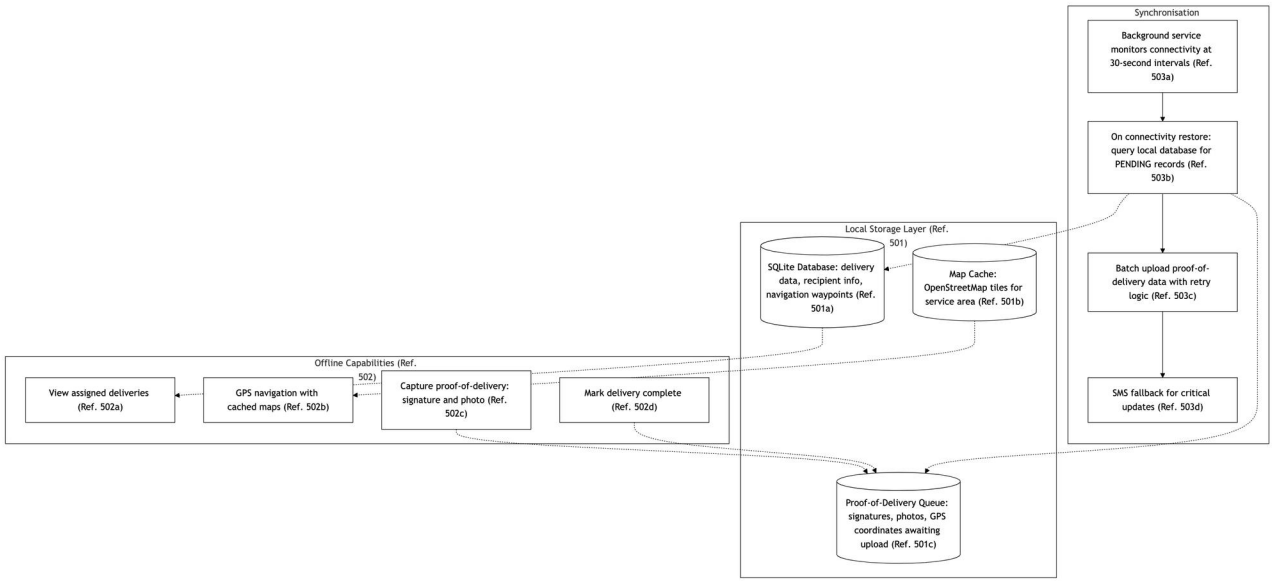
Synchronisation
Background service monitors connectivity at 30-second intervals (Ref. 503a)
On connectivity restore: query local database for PENDING records (Ref. 503b)
Batch upload proof-of-delivery data with retry logic (Ref. 503c)
SMS fallback for critical updates (Ref. 503d)
Local Storage Layer (Ref. 501)
SQLite Database: delivery data, recipient info, navigation waypoints (Ref. 501a)
Map Cache: OpenStreetMap tiles for service area (Ref. 501b)
Proof-of-Delivery Queue: signatures, photos, GPS coordinates awaiting upload (Ref. 501c)
Offline Capabilities (Ref. 502)
View assigned deliveries (Ref. 502a)
GPS navigation with cached maps (Ref. 502b)
Capture proof-of-delivery: signature and photo (Ref. 502c)
Mark delivery complete (Ref. 502d)

Fig. 5

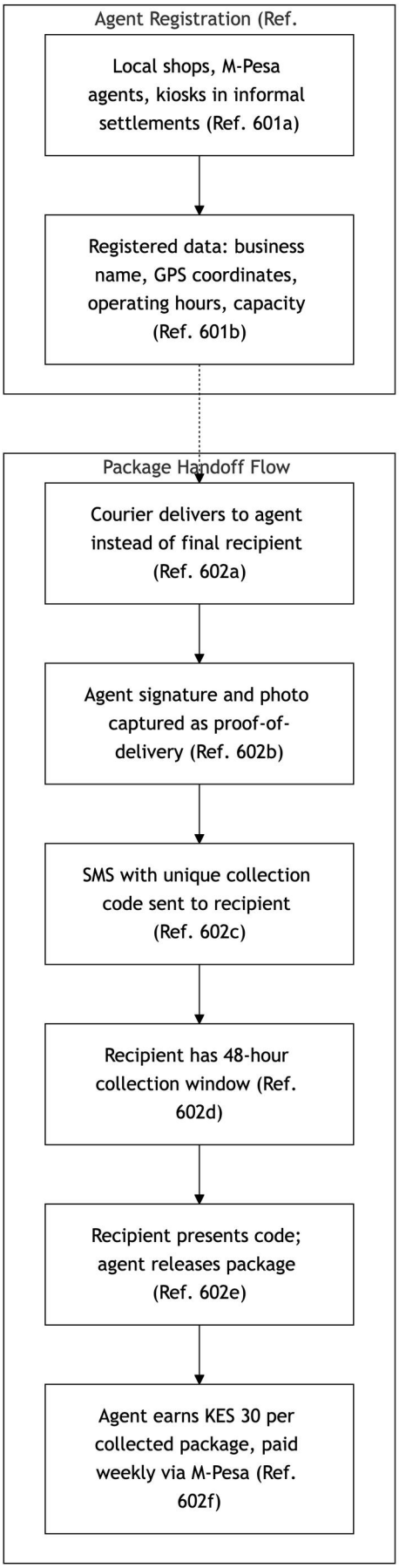
Agent Registration (Ref.
Local shops, M-Pesa agents, kiosks in informal settlements (Ref. 601a)
Registered data: business name, GPS coordinates, operating hours, capacity (Ref. 601b)
Package Handoff Flow
Courier delivers to agent instead of final recipient (Ref. 602a)
Agent signature and photo captured as proof-of-delivery (Ref. 602b)
SMS with unique collection code sent to recipient (Ref. 602c)
Recipient has 48-hour collection window (Ref. 602d)
Recipient presents code; agent releases package (Ref. 602e)
Agent earns KES 30 per collected package, paid weekly via M-Pesa (Ref. 602f)

Fig. 6